# GMT

## Equations and code architecture

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This manual describes the physical, numerical, acoustic, statistical and signal-processing equations in this distribution. Pixel layout and form validation are excluded. Illustrative parameters are not measurements.

References are relative to the repository root and use one-based lines. Ranges identify implementing function blocks; single lines identify code anchors. A compact JavaScript line may contain several equations. Code is authoritative for bounds and implementation details. source-map.json records anchors and SHA-256 hashes; verify_references.py detects changes.

Conventions: SI unless stated otherwise; $f$ in Hz, $\omega = 2\pi f$ in rad/s, sample rate $f_s$, time $t$ in seconds, $i^2 = -1$. Amplitude gain $10^{g/20}$ for g in dB; $\operatorname{sinc} x = \sin x/x$, with $\operatorname{sinc} 0 = 1$.

# Contents

**1 Models and equations** **3**

1.1 Orthotropic tensor . . . 3
1.2 TET10 and quadrature . . . 3
1.3 Elastic modes . . . 3
1.4 Reference outlines and cutaway . . . 3
1.5 Scaling and arch . . . 4
1.6 Soundhole and apertures . . . 4
1.7 SVG transforms and curves . . . 4
1.8 Local frames and losses . . . 5
1.9 Mechanical ports and surfaces . . . 5
1.10 Internal TET4 acoustics . . . 5
1.11 Static residual modes . . . 5
1.12 Aperture and Helmholtz . . . 6
1.13 Vibroacoustic solve . . . 6
1.14 Exterior radiation . . . 6
1.15 Corde : String tuning and loss . . . 6
1.16 Stiff-string impedance . . . 7
1.17 Coupled loading and passivity . . . 7
1.18 Blocked force and synthesis . . . 7
1.19 Phrases and tempo . . . 8
1.20 Monte Carlo dispersion . . . 8
1.21 FFT and spectral windows . . . 8
1.22 Peak, RMS and WAV . . . 8
1.23 Monitoring and band selection . . . 9
1.24 Display precision . . . 9
1.25 Universal modal interchange . . . 9

**2 File tree** **10**

**3 Dependencies** **11**

**4 Equation index** **13**

**5 Build and limitations** **14**

# 1 Models and equations

## 1.1 Orthotropic tensor

**Code:** `backend/catalog.py` : 61-65

$$\sigma = C\varepsilon, \quad C = S^{-1}, \quad S \succ 0, \tag{1}$$
$$S_{ii} = 1/E_i \ (i = 1, 2, 3), \quad (S_{44}, S_{55}, S_{66}) = (1/G_{LR}, 1/G_{LT}, 1/G_{RT}), \tag{2}$$
$$S_{12} = -\nu_{LR}/E_L, \quad S_{13} = -\nu_{LT}/E_L, \quad S_{23} = -\nu_{RT}/E_R, \tag{3}$$
$$S_{ij} = S_{ji}. \tag{4}$$

Voigt order is LL, RR, TT, LR, LT, RT with engineering shear. GPa converts to Pa by 1e9. Symmetric compliance enforces Poisson reciprocity. Nonpositive energy is rejected.

## 1.2 TET10 and quadrature

**Code:** `backend/fem.py` : 17-21

$$N_i = \lambda_i(2\lambda_i - 1), \quad N_{ij} = 4\lambda_i\lambda_j \ (i < j), \quad \sum_{i=1}^{4} \lambda_i = 1, \tag{5}$$
$$x = \sum_{a=1}^{10} N_a x_a, \quad J = \partial x/\partial \xi, \quad \nabla_x N = J^{-T}\nabla_\xi N, \tag{6}$$
$$K_e = \int B^T C B \, dV, \qquad M_e = \rho \int N^T N \otimes I_3 \, dV. \tag{7}$$

Four vertices and six edge midpoints. B computes engineering strain; mass is consistent. Eleven-point quadrature: centroid weight -74/5625; four permutations of (11/14,1/14,1/14,1/14) with weight 343/45000; six permutations of (a,a,b,b), where a=(1+sqrt(5/14))/4 and b=(1-sqrt(5/14))/4, with weight 28/1125. These are exact algorithm constants, not measured precision. Weights sum to 1/6.

## 1.3 Elastic modes

**Code:** `backend/fem.py` : 53-99

$$K\phi_j = \lambda_j M\phi_j, \quad \phi_i^T M\phi_j = \delta_{ij}, \tag{8}$$
$$f_j = \sqrt{\lambda_j}/(2\pi), \qquad r_j = \frac{\|K\phi_j - \lambda_j M\phi_j\|}{\max(\|K\phi_j\|, \epsilon)}. \tag{9}$$

Sparse shifted eigensolver, removal of constrained degrees and six free-free rigid modes. Normalization is mass-based. Low residual does not establish mesh convergence; out-of-band modes do not justify accurate extrapolation.

## 1.4 Reference outlines and cutaway

**Code:** `backend/contours.py` : 16-17, 28-57, 59-77

$$(x_j, y_j) = (W u_j, L v_j), \tag{10}$$
$$B(t) = (1-t)^3 P_0 + 3(1-t)^2 t P_1 + 3(1-t) t^2 P_2 + t^3 P_3, \tag{11}$$
$$b(t) = t^2(3-2t), \quad a\prime(t) = a(t) + (1-b)(N-a_0) + b(R-a_1). \tag{12}$$

Normalized (u,v) points are GMT 1.6 reference data documented in CONTOURS.md. Width W and length L in metres scale the outline. These are no longer original PCHIP half-widths. Cutaway adapts to neck N and shoulder R; depth follows a cosine weight ending at the horn, with a tangent Bézier join. Invalid shapes or area below 65% are rejected. Separate content conditions: PUBLICATION.md.

## 1.5 Scaling and arch

**Code:** `backend/contours.py` : 59-77; `backend/geometry.py` : 57-62, 64-67

$$x' = W[x - (x_{min} + x_{max})/2]/(x_{max} - x_{min}), \tag{13}$$
$$y' = L(y - y_{min})/(y_{max} - y_{min}), \tag{14}$$
$$a(x, y) = H \max[0, 1 - (2x/W)^2] \max[0, \sin(\pi y/L)]^{1.2}, \tag{15}$$
$$z' = z + t a_{top} - (1-t) a_{back}, \quad t = \mathrm{clip}[(z+d)/d, 0, 1]. \tag{16}$$

L, W, H and d are meters after mm input conversion. Imported outline is centered and scaled. Simplification is bounded by Hausdorff distance. Arch warps the mesh; this parametric model does not copy a historical plate.

## 1.6 Soundhole and apertures

**Code:** `backend/geometry.py` : 12-25, 27-36, 38-43

$$x(\theta) = c_x + r_x \cos\theta, \quad y(\theta) = c_y + r_y \sin\theta, \tag{17}$$
$$p_{SVG} = (width\, u + c_x, height\, v + c_y), \tag{18}$$
$$x_f(t) = c_x \pm [0.21W + 0.7 r_x \sin(2\pi t)], \quad y_f(t) = c_y + r_y(2t-1). \tag{19}$$

Width and height control soundhole size. Circle/ellipse uses 40 segments; D opening is clipped by a half-plane. f-holes are lines buffered by 0.32rx. SVG rings scale as separate apertures, checked against ribs, bridge and each other.

## 1.7 SVG transforms and curves

**Code:** `backend/svg_holes.py` : 147-189, 27-43, 45-55, 57-76

$$\widetilde{p}' = T_1 T_2 \cdots T_n \widetilde{p}, \quad \widetilde{p} = (x, y, 1)^T, \tag{20}$$
$$R(\phi) = \begin{pmatrix} \cos\phi & -\sin\phi \\ \sin\phi & \cos\phi \end{pmatrix}, \quad p' = R(p-c) + c, \tag{21}$$
$$p_{ij}^{new} = (p_i + p_{i+1})/2, \tag{22}$$
$$p(\theta) = c + R(\phi)(r_x \cos\theta, r_y \sin\theta)^T, \tag{23}$$
$$\Delta\theta = 2 \arccos[\mathrm{clip}(1 - tol/\max(r_x, r_y), -1, 1)]. \tag{24}$$

Affine matrices, rotations, translation, scale and skew compose in SVG order. Bézier uses de Casteljau subdivision until chord-distance and control-polygon excess meet tolerance, depth at most 16. Elliptic arcs use SVG endpoint-to-center conversion and sagitta-bounded step, at most 4096 points. Only inert geometry is read, with no script or external-link execution.

## 1.8 Local frames and losses

**Code:** `backend/fem.py` : 23-44, 46-51, 53-99

$$\varepsilon_{LRT} = R^T \varepsilon R, \quad e_{LRT} = Qe, \quad C_{global} = Q^T C Q, \tag{25}$$

$$K_{loss} = \sum_p \eta_p K_p, \quad \eta_j = \phi_j^T K_{loss} \phi_j / \omega_j^2, \quad \zeta_j = \eta_j / 2. \tag{26}$$

Material frame follows each component, grain/ring angles and arch, with orthonormalization. Loss matrix sums component stiffnesses weighted by loss; modal damping projects onto mass-normalized modes.

## 1.9 Mechanical ports and surfaces

**Code:** `backend/fem.py` : 102-123, 125-139; `backend/response.py` : 12-23

$$N_i^{\triangle} = \lambda_i(2\lambda_i - 1), \quad N_{ij}^{\triangle} = 4\lambda_i\lambda_j, \tag{27}$$

$$u_{port} = B^T u, \quad F_{nodes} = B F_{port}, \quad P = \Phi^T B, \tag{28}$$

$$S_{face,j} = A_{face}\, n \cdot \frac{u_{mid1,j} + u_{mid2,j} + u_{mid3,j}}{3}. \tag{29}$$

Twelve ports represent six strings in normal and tangential directions. Interpolation and force spreading use the same quadratic functions. Exact quadratic triangle area averaging gives zero vertex weights and one third at edge midpoints, yielding radiating panel volume displacement.

## 1.10 Internal TET4 acoustics

**Code:** `backend/fem.py` : 141-153, 155-166

$$K_e^a = V_e \nabla N_i \cdot \nabla N_j, \quad M_e^a = V_e(\mathbf{1} + I)/20, \tag{30}$$

$$K^a \psi_j = k_j^2 M^a \psi_j, \quad \omega_j^a = c k_j, \quad \psi_i^T M^a \psi_j = \delta_{ij}, \tag{31}$$

$$C_{ja} = \int_\Gamma \psi_a (n \cdot \phi_j)\, dS, \quad h_a = \langle \psi_a \rangle_{hole}. \tag{32}$$

Pressure uses linear functions and structure quadratic ones. Surface coupling uses six-point triangle quadrature. Acoustic modes, including uniform mode, interact with structure and aperture coordinates.

## 1.11 Static residual modes

**Code:** `backend/reduction.py` : 11-46

$$KX = B, \quad R = X - \Phi \Omega^{-2} \Phi^T B, \tag{33}$$

$$R \leftarrow R - \Phi \Phi^T M R, \quad R^T K R = U \Lambda U^T, \quad Q = R U \Lambda^{-1/2}, \tag{34}$$

$$Q^T K Q z = \lambda Q^T M Q z, \tag{35}$$

$$\epsilon_{stat} = \frac{\| P^T \Omega^{-2} P + P_r^T \Lambda_r^{-1} P_r - B^T X \|}{\| B^T X \|}. \tag{36}$$

Free-free static solve uses a saddle system with six mass-orthogonal rigid motions. Orthogonalization repeats twice and small energy eigenvalues are removed. Residual basis completes port response but does not make the model exact at arbitrarily high frequency.

## 1.12 Aperture and Helmholtz

**Code:** backend/response.py : 35-68, 82-91

$$r_{eq} = \sqrt{S/\pi}, \quad \ell_{end} = a_{end} r_{eq}, \quad M_h = \rho_a \ell_{end}/S, \tag{37}$$

$$R_h = r_{factor}\left[\frac{\rho_a \omega^2}{4\pi c} + 0.05 M_h \max\left(1, c\sqrt{S/(V\ell_{end})}\right)\right], \tag{38}$$

$$d_h = -\omega^2 M_h + i\omega R_h, \tag{39}$$

$$f_H^{display} = \frac{c}{2\pi}\sqrt{\frac{S}{V(t_{top} + a_{end} r_{eq})}}. \tag{40}$$

The solver uses aperture mass based on end correction and approximate resistance. The displayed Helmholtz guide adds top thickness and assumes a rigid cavity; it does not replace the coupled solution. These two effective lengths must not be conflated.

## 1.13 Vibroacoustic solve

**Code:** backend/response.py : 25-33, 70-80

$$d_j = \omega_j^2 - \omega^2 + 2i\omega\zeta_j\omega_j, \tag{41}$$

$$d_a = k_a^2 - (\omega/c)^2 + 2i\zeta_a\omega\omega_a/c^2, \quad \zeta_a = 0.025, \tag{42}$$

$$D = \mathrm{diag}(d_j, d_h), \quad A = \begin{bmatrix} C \\ h \end{bmatrix}, \quad q_0 = D^{-1}F, \tag{43}$$

$$W = \mathrm{diag}(d_a/(\rho_a\omega^2)) - A^T D^{-1} A, \tag{44}$$

$$a = W^{-1}A^T q_0, \quad q = q_0 + D^{-1}Aa, \quad Y = i\omega P^T q. \tag{45}$$

This Schur complement eliminates mechanical variables in reduced space, then recovers structure and aperture coordinates. Peak edits multiply both emitting and receiving projections by 10^(gain/40). DC is treated separately.

## 1.14 Exterior radiation

**Code:** backend/response.py : 35-68, 70-80

$$G(d, \omega) = e^{-i\omega d/c}/d, \tag{46}$$

$$H_{pF} = -\frac{\rho_a \omega^2}{4\pi}\left[\sum_s G(d_s, \omega) S_s q_s + G(d_h, \omega) q_h\right]. \tag{47}$$

Monopole-panel sum propagates toward the listening point. Distance is bounded at 3 cm. Exterior radiation is approximate, not full BEM with radiation-loading feedback. Simulated pressure is not microphone calibration.

## 1.15 Corde : String tuning and loss

**Code:** backend/strings.py : 10-13, 5-8

$$L = L_0 2^{-r/12}, \quad f_o = f_{A4} 2^{(m-69)/12}, \tag{48}$$
$$T = 4\mu L_0^2 f_o^2 - \pi^2 EI/L_0^2, \tag{49}$$
$$f_0 = \frac{1}{2\pi}\sqrt{[T(\pi/L)^2 + EI(\pi/L)^4]/\mu}, \tag{50}$$
$$\alpha(f) = loss + hf\, f + 2\,10^{-8} f^2 + palm(12 + 0.015f). \tag{51}$$

Mu is entered in g/m and converted to kg/m. Negative tension is rejected. Palm losses are phenomenological, not a nonlinear hand-string contact solution.

## 1.16 Stiff-string impedance

**Code:** `backend/strings.py` : 15-20

$$\omega_c = \omega - i\alpha, \quad \Delta = \sqrt{T^2 + 4EI\mu\omega_c^2}, \tag{52}$$
$$k^2 = \frac{2\mu\omega_c^2}{T+\Delta}, \quad \kappa^2 = \frac{T+\Delta}{2EI}, \quad A = \frac{\kappa^2}{\kappa^2 + k^2}, \quad B = 1 - A, \tag{53}$$
$$D_s = (T + EIk^2)k\cot(kL)A - EIk^2\kappa\coth(\kappa L)B, \tag{54}$$
$$Z_s = D_s/(i\omega). \tag{55}$$

The $k^2$ form avoids ill-conditioned subtraction. Negligible EI gives D=T k cot(kL), k=omega_c sqrt(mu/T). Code sets Z=0 at DC. This passive impedance represents bridge string loading.

## 1.17 Coupled loading and passivity

**Code:** `backend/response.py` : 82-91; `backend/audio.py` : 27-67

$$Q = [I + \mathrm{diag}(Z_s)Y]^{-1}, \quad Y_{loaded} = YQ, \quad H_{loaded} = HQ, \tag{56}$$
$$F = QF_{blocked}, \quad v = YF, \quad p = HF, \tag{57}$$
$$Y_H = (Y + Y^*)/2, \quad \lambda_{min}(Y_H) \geq 0 \text{ for passivity.} \tag{58}$$

String impedances load twelve ports; unexcited strings can remain active for sympathy. Passivity is sampled on the frequency grid, not proven continuously at all frequencies.

## 1.18 Blocked force and synthesis

**Code:** `backend/audio.py` : 27-67, 69-75, 8-12

$$q_n = \frac{2F\sin(k_n x_p)}{L(Tk_n^2 + EIk_n^4)} s_n^{exc}, \tag{59}$$
$$F_b(t) = \sum_n (Tk_n + EIk_n^3) q_n \cos(\omega_n t) e^{-\alpha_n t}, \tag{60}$$
$$s_n^{exc} = \mathrm{sinc}(k_n w/2) e^{-(1-hardness)f_n/f_h}, \tag{61}$$
$$(F_z, F_x) = F_b(\cos\theta, \sin\theta), \quad N = \mathrm{nextfast}(N_{audio} + 2f_s). \tag{62}$$

Blocked force comes from a fixed-end plucked string; loaded solve gives actual force, velocity and pressure. FFT adds two seconds of guard; transfer real and imaginary parts are interpolated. Beyond the grid decay is fmax/f. Band edges are tapered; finite-duration frequency-domain synthesis remains a transient approximation.

## 1.19 Phrases and tempo

**Code:** `backend/repertoire.py` : 11-35; `web/repertoire.js` : 15

$$t_e = 60 b_e/\mathrm{BPM}, \quad L_{fret} = L_0 2^{-fret/12}, \tag{63}$$

$$y_j = \sum_e strength_e \, x_{string_e, fret_e}[j - \mathrm{round}(t_e f_s)] \, w_e[j]. \tag{64}$$

Notes are grouped by string/fret to bound memory. Each attack is an independent linear response using current guitar settings. Note release receives a 15 ms fade. Finger changes do not model continuously time-varying sympathetic interaction.

## 1.20 Monte Carlo dispersion

**Code:** `backend/uncertainty.py` : 7-39

$$\sigma = \sqrt{\ln(1 + CV^2)}, \quad z = \sqrt{r} z_c + \sqrt{1 - r} z_i, \tag{65}$$

$$a = \exp(\sigma z - \sigma^2/2), \quad g_{dB} = 40 \log_{10} a, \tag{66}$$

$$(q_{10}, q_{50}, q_{90}) = \mathrm{quantile}_{0.1,0.5,0.9} \, |H^{(s)}(f)|. \tag{67}$$

Lognormal factors have unit mean. Correlation r shares a common draw within each parameter family. Frequencies, losses and gains are perturbed with bounds. Pointwise quantiles represent chosen dispersion, not experimental confidence intervals; mode shapes, air and residual basis remain fixed.

## 1.21 FFT and spectral windows

**Code:** `web/signal.js` : 2

$$X_k = \sum_{j=0}^{N-1} w_j x_j e^{-2\pi i j k/N}, \quad A_k = c_k |X_k| / \sum_j w_j, \tag{68}$$

$$L_k = 20 \log_{10} \max(10^{-14}, A_k), \quad a = 2\pi j/(N-1), \tag{69}$$

$$w_H = 0.5 - 0.5 \cos a, \quad w_{Ham} = 0.54 - 0.46 \cos a, \tag{70}$$

$$w_{Blackman} = 0.42 - 0.5 \cos a + 0.08 \cos 2a, \quad \Delta f = f_s/N. \tag{71}$$

Radix-2 FFT requires power-of-two size. Spectrum retains N/2 bins; ck is 1 at DC and 2 elsewhere. Rectangular window is one. Zero padding does not create additional physical resolution.

## 1.22 Peak, RMS and WAV

**Code:** `web/signal.js` : 22

$$P = \max_j |x_j|, \quad x_{RMS} = \sqrt{N^{-1} \sum_j x_j^2}, \tag{72}$$

$$q_{16} = \mathrm{round}(s \ \mathrm{clip}(x, -1, 1)), \quad s \in \{32767, 32768\}, \tag{73}$$

$$bytes_{data} = N \, channels \, (bits/8), \quad byteRate = f_s \, channels \, (bits/8). \tag{74}$$

PCM clips samples to [-1,1]. Variants using 32768 for negative values use the full signed range; GMT/ToneWood use 32767 on both sides. DataView conversion without explicit rounding truncates. Floating WAV, where offered, retains physical quantities whose units must be read with metadata.

## 1.23 Monitoring and band selection

**Code:** `web/signal.js` : 23, 32, 41

$$g_{req} = v10^{g_{dB}/20} \begin{cases} 0.8/P & \text{peak} \\ 0.12/RMS & \text{RMS} \\ 1 & \text{physical} \end{cases}, \tag{75}$$

$$g = \min(g_{req}, 0.9/P), \tag{76}$$

$$w(f) = \tfrac{1}{2}[1 + \cos(\pi d(f)/\Delta)] \quad (0 < d < \Delta). \tag{77}$$

Listening gain is peak-capped. Time-frequency selection uses FFT with 30 ms padding, 10-100 Hz transition width and 5 ms time fades. Graphical min/max envelopes retain block extrema without filtering physical data.

## 1.24 Display precision

**Code:** `web/suite.js` : 13, 19

$$d = \lfloor \log_{10} u \rfloor - 1, \quad h = 10^d, \tag{78}$$

$$x_{display} = h\,\mathrm{round}(x/h), \quad u = \max(x - l, r - x). \tag{79}$$

Display only: generally three significant digits, at most four for dimensions and frequencies. For FEMU interval [l,r], the larger half-width u determines the shared decimal place of the estimate and bounds (two significant digits in width). Calculations and exports are not rounded. Invalid intervals use standard significant formatting. Exact algorithm constants and line numbers are not measurements and retain full digits.

## 1.25 Universal modal interchange

**Code:** `backend/universal.py` : 37-43, 6-35

$$f_j = \omega_j/(2\pi), \quad \zeta_j = \eta_j/2, \tag{80}$$

$$\Phi_j^T M \Phi_j = 1, \quad n_{UFF} = n_{local} + 1. \tag{81}$$

UNV and UFF are two extensions of the same ASCII Universal format. Original dry FE modes are exported, without virtual modal edits or acoustic modes. Dataset 164: SI units; 2411: all solid nodes and XYZ coordinates in metres; 2412: triangular visualization skin, not a volume FE mesh; 55: real mass-normalized XYZ vectors, frequencies in Hz, equivalent viscous damping eta/2. Local node IDs are renumbered 1 through N. Dataset 55 standard E13.5 records retain six significant digits; NPZ keeps the complete numerical arrays.

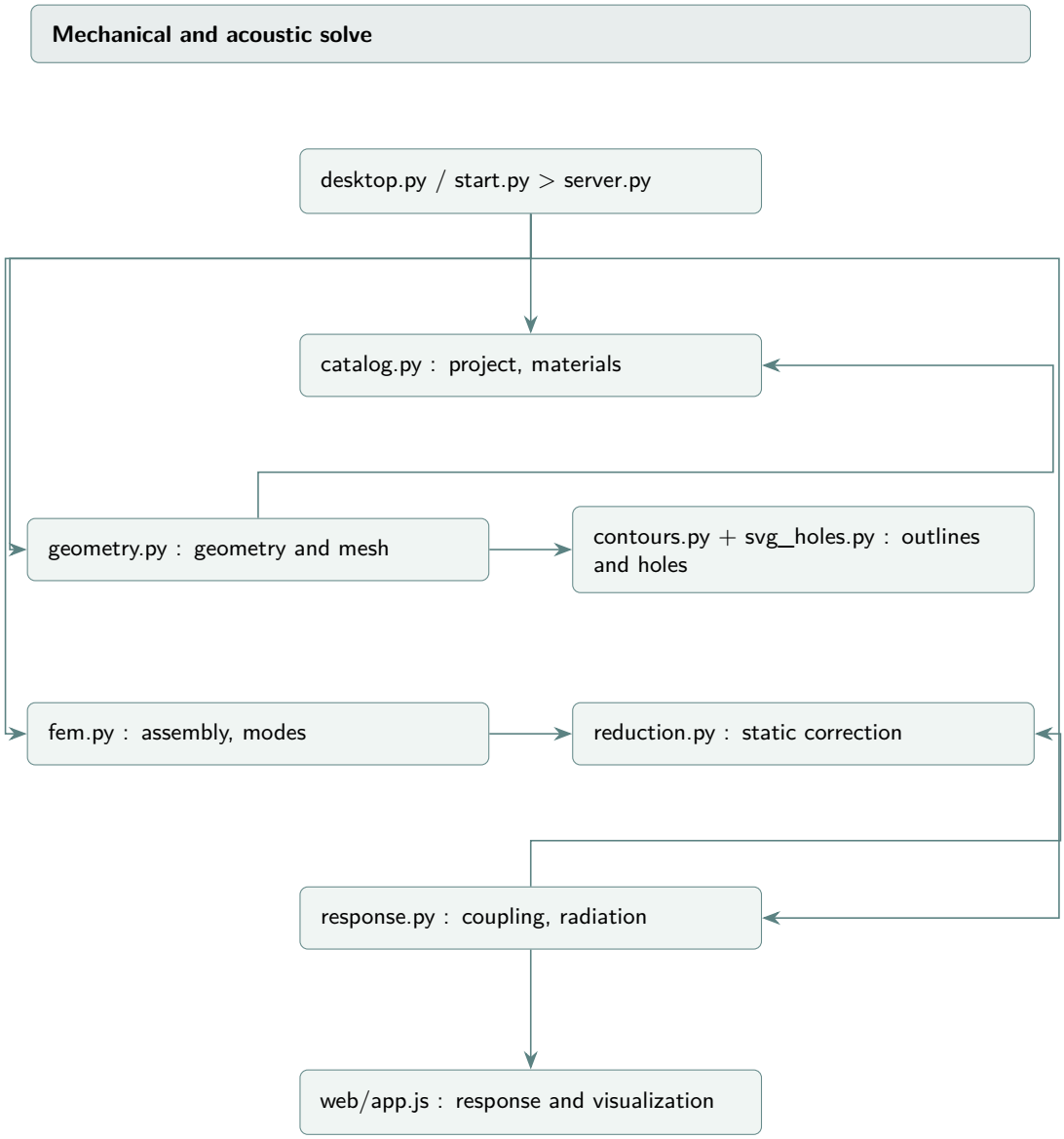
## 2 File tree

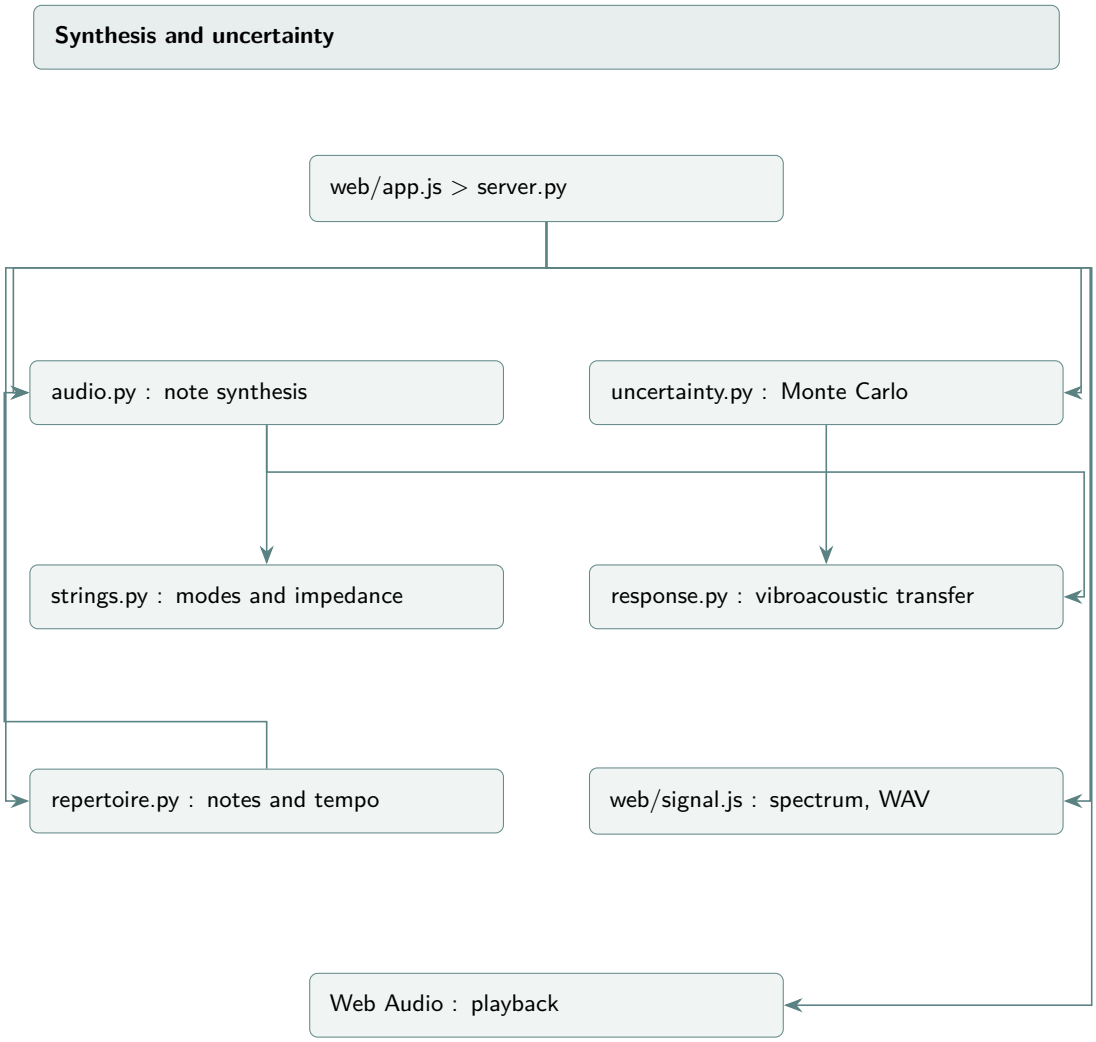
File groups describe active modules. The repository CSV manifest lists every distributed file.



# 3 Dependencies

Arrows: calls, imports or data flow. Subsystems are separated to keep boxes and links readable.




Synthesis and uncertainty
web/app.js > server.py
audio.py : note synthesis
uncertainty.py : Monte Carlo
strings.py : modes and impedance
response.py : vibroacoustic transfer
repertoire.py : notes and tempo
web/signal.js : spectrum, WAV
Web Audio : playback

# 4 Equation index

| N° | Group | File, line |
|---|---|---|
| 1 | Orthotropic tensor | `backend/catalog.py` : 61-65 |
| 2 | TET10 and quadrature | `backend/fem.py` : 17-21 |
| 3 | Elastic modes | `backend/fem.py` : 53-99 |
| 4 | Reference outlines and cutaway | `backend/contours.py` : 59-77 |
| 4 | Reference outlines and cutaway | `backend/contours.py` : 16-17 |
| 4 | Reference outlines and cutaway | `backend/contours.py` : 28-57 |
| 5 | Scaling and arch | `backend/contours.py` : 59-77 |
| 5 | Scaling and arch | `backend/geometry.py` : 57-62 |
| 5 | Scaling and arch | `backend/geometry.py` : 64-67 |
| 6 | Soundhole and apertures | `backend/geometry.py` : 12-25 |
| 6 | Soundhole and apertures | `backend/geometry.py` : 27-36 |
| 6 | Soundhole and apertures | `backend/geometry.py` : 38-43 |
| 7 | SVG transforms and curves | `backend/svg_holes.py` : 27-43 |
| 7 | SVG transforms and curves | `backend/svg_holes.py` : 45-55 |
| 7 | SVG transforms and curves | `backend/svg_holes.py` : 57-76 |
| 7 | SVG transforms and curves | `backend/svg_holes.py` : 147-189 |
| 8 | Local frames and losses | `backend/fem.py` : 23-44 |
| 8 | Local frames and losses | `backend/fem.py` : 46-51 |
| 8 | Local frames and losses | `backend/fem.py` : 53-99 |
| 9 | Mechanical ports and surfaces | `backend/fem.py` : 125-139 |
| 9 | Mechanical ports and surfaces | `backend/fem.py` : 102-123 |
| 9 | Mechanical ports and surfaces | `backend/response.py` : 12-23 |
| 10 | Internal TET4 acoustics | `backend/fem.py` : 141-153 |
| 10 | Internal TET4 acoustics | `backend/fem.py` : 155-166 |
| 11 | Static residual modes | `backend/reduction.py` : 11-46 |
| 12 | Aperture and Helmholtz | `backend/response.py` : 35-68 |
| 12 | Aperture and Helmholtz | `backend/response.py` : 82-91 |
| 13 | Vibroacoustic solve | `backend/response.py` : 25-33 |
| 13 | Vibroacoustic solve | `backend/response.py` : 70-80 |
| 14 | Exterior radiation | `backend/response.py` : 35-68 |
| 14 | Exterior radiation | `backend/response.py` : 70-80 |
| 15 | Corde : String tuning and loss | `backend/strings.py` : 5-8 |
| 15 | Corde : String tuning and loss | `backend/strings.py` : 10-13 |
| 16 | Stiff-string impedance | `backend/strings.py` : 15-20 |
| 17 | Coupled loading and passivity | `backend/response.py` : 82-91 |
| 17 | Coupled loading and passivity | `backend/audio.py` : 27-67 |
| 18 | Blocked force and synthesis | `backend/audio.py` : 8-12 |
| 18 | Blocked force and synthesis | `backend/audio.py` : 27-67 |
| 18 | Blocked force and synthesis | `backend/audio.py` : 69-75 |
| 19 | Phrases and tempo | `backend/repertoire.py` : 11-35 |
| 19 | Phrases and tempo | `web/repertoire.js` : 15 |
| 20 | Monte Carlo dispersion | `backend/uncertainty.py` : 7-39 |
| 21 | FFT and spectral windows | `web/signal.js` : 2 |
| 22 | Peak, RMS and WAV | `web/signal.js` : 22 |
| 23 | Monitoring and band selection | `web/signal.js` : 23 |
| 23 | Monitoring and band selection | `web/signal.js` : 41 |
| 23 | Monitoring and band selection | `web/signal.js` : 32 |
| 24 | Display precision | `web/suite.js` : 13 |

| N° | Group | File, line |
|---|---|---|
| 24 | Display precision | `web/suite.js` : 19 |
| 25 | Universal modal interchange | `backend/universal.py` : 6-35 |
| 25 | Universal modal interchange | `backend/universal.py` : 37-43 |

# 5 Build and limitations

From docs/, run pdflatex -halt-on-error manual.tex twice. The *-standalone.tex wrappers compile each diagram separately. A4 paper, 21 mm margins, no automatic diagram shrinkage. Third-party sources, licenses and credits are in CREDITS.md and SOURCES.md. This documents implementation, not experimental validation.